

University of Armed Forces - ESPE

Ostin Torres
Josselyn Ulcuango
Mateo Unda
Yandry Vélez

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Object Oriented Programming
PhD Edison Lascano

Project: LegalSync — Legal Case Management System

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1. INTRODUCTION

1.1. Purpose

The purpose of this document is to specify the functional and non-functional requirements of LegalSync, a software solution designed to manage legal cases for a law firm.

This document establishes the system's required functionalities, the users who will interact with it, the constraints it must respect, and the conditions used to verify compliance with its requirements.

This specification will serve as a reference for the analysis, design, implementation, and testing phases of the project. It will also help development team members and stakeholders develop a common understanding of the product's scope.

1.2. Product Scope

LegalSync aims to centralize information related to clients, defendants, court cases, assigned lawyers, documents, evidence, and legal deadlines.

Currently, case management is performed using spreadsheets and separate files. This process leads to duplicated work, difficulties locating information, update errors, and problems identifying the users responsible for changes.

The proposed system will allow users to register and update case files, associate them with clients and defendants, attach references to documents, retrieve information through searches, monitor legal deadlines, record modifications, and generate text documents from previously stored information.

The system will also support authentication through usernames and passwords, role-based permission management, and data import from CSV files.

The main objective is to improve the organization, traceability, and efficiency of the firm's administrative processes while reducing dependence on spreadsheets and manual records.

1.3. Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirements Specification.
LegalSync	Name of the legal case management system.
CodeLex	Name of the project development team.
LegalCase	A record containing information about a legal proceeding.
Case File	A collection of data, documents, and evidence associated with a case.
Client	A person receiving legal services from the law firm.
Defendant	A person against whom a lawsuit or legal action is brought.
Lawyer	A legal professional responsible for managing one or more cases.
Legal Assistant	A user who supports administrative tasks and case file updates.
Partner	A user with administrative privileges and the ability to manage permissions.

Functional Requirement (FR)	A function or behavior that the system must provide
Non-Functional Requirement (NFR)	A quality, security, performance, or operational condition that the system must satisfy.
UML	Unified Modeling Language, used to represent the system design.
CSV	Comma-Separated Values file format used for data exchange.
TXT	Plain-text file format.
PDF	Portable Document Format.
AuditLog	A chronological record of actions or modifications performed.
Backlog	A prioritized list of features and requirements to be developed.

1.4. Project References

This specification is based on the following materials provided for the project:

1. HW04 — Definition: Problem definition, background, general overview, and scope of the LegalSync system.
2. HW05 — Requirement: Product Backlog PB-001 through PB-009, including priorities and acceptance criteria.
3. usecasediagram.vpp: UML use case diagram for the LegalSync system.

4. classesLegalSync (1).vpp: UML class diagram representing the main entities and their operations.

1.5. Document Conventions

Requirements will be identified using unique codes to facilitate tracking:

- FR-XXX: Functional Requirement.
- NFR-XXX: Non-Functional Requirement.
- UC-XXX: Use Case.
- DATA-XXX: Data Requirement.
- AC-XXX: Acceptance Criterion.

Each requirement will be written in a verifiable form. The word “shall” will be used to indicate mandatory behavior within the specified scope.

Priorities will be interpreted as follows:

- High: Essential functionality for core operations.
- Medium: Important functionality that complements system operation.
- Low: Additional functionality that may be implemented after higher-priority requirements.

2. OVERALL DESCRIPTION

2.1. Product Perspective

LegalSync is intended to centralize the management of legal case files that are currently distributed across spreadsheets, physical folders, and separate files.

The system will consolidate case information and allow users to retrieve related data from a single platform. Each case file may contain client information, defendant information, the assigned lawyer, the procedural status, related documents and evidence, and deadlines that must be addressed.

The application will include modules for authentication, user management, case registration, document references, deadline alerts, searching, auditing, data import, and document generation.

In the academic version defined by the Product Backlog, certain operations will be implemented using local files, in-memory collections, and a console or basic user interface. The overall product vision includes future development into a multiplatform solution with secure remote access and mobile notifications.

Therefore, this SRS distinguishes the functionalities defined for the academic implementation from the broader capabilities described in the product vision.

2.2. Problem Description

Manual legal case management presents the following problems:

1. Disorganized information: Case data and related documents may be stored in separate files.
2. Time loss: Manual updates and searches require additional administrative work.
3. Risk of errors: Information may become incomplete, outdated, or duplicated.
4. Lack of traceability: There is no sufficiently structured record identifying who modified each case file.
5. Risk of missed deadlines: Spreadsheets do not independently provide automated alerts for legal dates.
6. Limited access control: Different users may access information without adequate permission separation.
7. Difficult document retrieval: Locating evidence and files associated with a case may require checking multiple locations.

LegalSync aims to address these problems by centralizing records and automating priority administrative tasks.

2.3. System Objectives

2.3.1. General Objective

To develop a legal case management system that centralizes, organizes, retrieves, and updates a law firm's legal information while incorporating user authentication, permission control, deadline tracking, and modification traceability.

2.3.2. Specific Objectives

- Register and update case files containing mandatory fields.
- Manage users and their permissions according to assigned roles.
- Associate file references and evidence with the corresponding cases.
- Calculate the number of days remaining until legal deadlines and display alerts.
- Facilitate case searches, primarily by client name.
- Record the date, time, and user responsible for modifications.
- Import structured information from CSV files.
- Generate TXT documents using case information.
- Reduce errors associated with manual information management.

2.4. User Classes and Characteristics

The use case diagram identifies four main actors: Partner, Lawyer, Junior Lawyer, and Legal Assistant.

Actor	Description and Responsibilities
Partner	Manages accounts, assigns roles, controls permissions, and supervises authorized records and operations.
Lawyer	Retrieves and manages assigned case files, reviews legal deadlines, and accesses related information.
Junior Lawyer	Participates in case management according to assigned permissions.
Legal Assistant	Supports data entry and updates, document organization, and authorized administrative tasks.

The final permissions for each role must be configured according to the rules established by the partners. The source documents explicitly specify role-based access control and the partners' ability to restrict functions, but they do not provide a complete permission matrix for every operation.

2.5. Operating Environment

According to the academic requirements and product vision, the following operating environment is considered:

- An application designed for legal case management.
- Interaction through a console or user interface.
- Reading and writing local text files.
- CSV files for importing data.
- References to PDF and Word files associated with case files.
- Audit records stored in a local file named log.txt.
- Access through user credentials.
- A conceptual design intended to support future use on computers, tablets, and mobile phones.

The source documents do not specify the final operating system, database engine, mandatory programming language, or notification service provider. These elements must be defined during the technical design phase.

2.6. Product Constraints

1. The system shall validate mandatory fields before registering or updating a case file.
2. Initial authentication shall use credentials stored in a local text file, as specified in the Product Backlog.
3. Document references within the academic scope shall be stored as text file paths.
4. Initial alerts shall be displayed when the system starts, through the console or user interface.
5. Initial data import shall use CSV files.
6. Audit records shall be written to a local log.txt file.
7. The primary document generation functionality within the academic scope shall produce TXT files.
8. Administrative operations shall respect the permissions assigned to each role.

These constraints describe the academic implementation defined in HW05. They should not be interpreted as confirmation that all capabilities of a future multiplatform version have already been developed.

2.7. Assumptions and Dependencies

- Users will have valid credentials to access the system.
- Initial data may come from spreadsheets or CSV files.
- Files linked to cases must exist in a location accessible to the execution environment.
- The user responsible for registering a case must have the minimum required information.
- Legal dates must be entered correctly for the remaining-days calculation to be reliable.
- Document generation will depend on the availability of the fields used by the templates.
- External alerts or push notifications will depend on additional components not defined in the academic Product Backlog.

3. FUNCTIONAL REQUIREMENTS

Functional requirements describe the operations LegalSync shall perform. They are organized according to the nine functionalities in the HW05 Product Backlog, preserving their priorities and acceptance criteria.

3.1. Authentication and Role Management

Identification: FR-001 Name: Authentication and User Management Priority: High Source: PB-001 and PB-007.

Description:

The system shall allow authorized users to log in using credentials and enable partners to manage user accounts and permissions.

Specific Requirements:

- FR-001.1. The system shall allow user accounts to be created.
- FR-001.2. Each user shall have a login identifier and a password.
- FR-001.3. The system shall validate credentials before granting access.
- FR-001.4. The entered password shall be visually hidden during typing, using asterisks or another masking mechanism.
- FR-001.5. The system shall support the assignment of Partner, Lawyer, Junior Lawyer, and Legal Assistant roles.
- FR-001.6. The system shall restrict operations according to the permissions assigned to each user.
- FR-001.7. Partners shall be able to manage other users' permissions.
- FR-001.8. The system shall allow users to log out.
- FR-001.9. The system shall reject access when credentials are invalid.
- FR-001.10. The system shall allow users to change their passwords, according to the operations included in the class diagram.

Acceptance Criteria:

1. A user with valid credentials can log in.
2. A user with incorrect credentials cannot access protected functions.
3. The password is not displayed directly while being entered.
4. A user cannot perform an operation for which they lack permission.
5. A partner can manage accounts and permissions according to the available functions.

Security Note: The Product Backlog specifies that credentials are initially stored in a local text file. This is an academic implementation constraint, not a recommended practice for a production legal system. In a real-world version, passwords shall be stored using a secure cryptographic password-hashing mechanism rather than in plain text.

3.2. Case Registration and Updating

Identification: FR-002 Name: Legal Case Management Priority: High Source: PB-002.

Description:

The system shall allow users to register new case files and update information in existing cases while ensuring that all mandatory fields are completed.

Specific Requirements:

- FR-002.1. The system shall allow users to register a new legal case.
- FR-002.2. Each case shall have a unique identifier.
- FR-002.3. The case file shall include, at a minimum, the case number, title, status, opening date, client, defendant, and assigned lawyer.
- FR-002.4. The system shall validate mandatory fields before saving a case file.
- FR-002.5. The system shall allow users to update information in an existing case.
- FR-002.6. The system shall allow users to view case details.
- FR-002.7. The system shall associate documents, evidence, and deadlines with the corresponding case.
- FR-002.8. The system shall prevent a case from being registered when mandatory information is missing.
- FR-002.9. Modifications shall respect the user's permissions.
- FR-002.10. All modifications shall be recorded through the defined auditing mechanism.

Main Case File Data:

Field	Description
caseId	Case identifier.
docketNumber	Court case number.
title	Case title or description.
status	Current case status.
openingDate	Date the case was opened
client	Associated client.
defendant	Associated defendant.
assignedLawyer	Lawyer responsible for the case.
documents	Related documents.
evidences	Related evidence.

deadlines	Associated deadlines

Acceptance Criteria:

1. An authorized user can register a case with complete information.
2. The system rejects registration when mandatory fields are missing.
3. An authorized user can modify an existing case file.
4. Case data remains available for subsequent retrieval.
5. Modifications are reflected in the audit log.

3.3. Document Reference Registration

Identification: FR-003 Name: Association of Document and Evidence References Priority: High Source: PB-003.

Description:

The system shall allow users to register file paths and associate them with the corresponding legal case files.

Specific Requirements:

- FR-003.1. The system shall allow users to register a document reference.
- FR-003.2. Each reference shall be associated with a case file.
- FR-003.3. The system shall store the file path as a text attribute.
- FR-003.4. The system shall allow references to PDF and Word documents to be registered.
- FR-003.5. The system shall support references to evidence related to cases.
- FR-003.6. The system shall allow stored paths to be retrieved from the case file.
- FR-003.7. The academic implementation shall not process or interpret the file's contents to satisfy the file-path registration requirement.
- FR-003.8. The system shall preserve the association between the document reference and its corresponding case file.

Example of a Reference:

C:\case1.pdf

The path above is an example of the format described in the Product Backlog; it does not represent an actual system file.

Acceptance Criteria:

1. An authorized user can register a file path.

2. The path is associated with the selected case file.
3. The path can be retrieved later.
4. The system stores the reference as text without analyzing the file's contents.

Scope Clarification: The general description in HW04 proposes uploading and previewing PDF and Word files. However, PB-003 specifies that the academic functionality consists of storing the path as text. Actual document uploading and viewing shall be considered an extension if they are not implemented separately.

3.4. Legal Deadline Management and Alerts

Identification: FR-004 Name: Legal Date Tracking Priority: High Source: PB-004.

Description:

The system shall register deadlines associated with case files, calculate the remaining time, and display visual alerts when a legal date is approaching.

Specific Requirements:

- FR-004.1. The system shall allow users to register a legal deadline.
- FR-004.2. Each deadline shall be associated with a case file.
- FR-004.3. Each deadline shall include a description, type, target date, number of days before the deadline for alerting, and completion status.
- FR-004.4. The system shall calculate the remaining days by comparing the target date with the current date.
- FR-004.5. The system shall determine whether a deadline is approaching according to the configured alert period.
- FR-004.6. The system shall display visual alerts when deadlines are approaching.
- FR-004.7. Alerts shall be displayed when the system starts, according to the Product Backlog acceptance criterion.
- FR-004.8. The system shall allow users to view alerts associated with deadlines.
- FR-004.9. The system shall allow a deadline to be marked as completed.
- FR-004.10. The system shall distinguish between pending, approaching, overdue, and completed deadlines when this status logic is implemented.

Calculation Rule:

Remaining Days = Target Date − Current Date.

The difference shall be calculated in calendar days unless the team establishes a different rule for judicial business days.

Acceptance Criteria:

1. The system allows a legal date to be registered.

2. The system correctly calculates the remaining days.
3. An alert is displayed when a deadline falls within the configured advance-notice period.
4. A completed deadline is not displayed as pending.
5. Deadlines can be retrieved in relation to their associated case file.

Identified Limitation: The Product Backlog requires local visual alerts when the system starts. Push notifications to mobile phones, described in HW04, constitute an additional capability requiring further specification and implementation.

3.5. Global Search by Client

Identification: FR-005 Name: Case Search Priority: High Source: PB-005.

Description:

The system shall allow users to locate case files using text-based search criteria, primarily the client's name.

Specific Requirements:

- FR-005.1. The system shall provide a case search function.
- FR-005.2. The search shall primarily use the client's name.
- FR-005.3. The system shall search stored data structures, such as arrays or lists, as defined in the implementation.
- FR-005.4. The system shall display records matching the search criterion.
- FR-005.5. The results shall allow users to identify the corresponding case file.
- FR-005.6. Authorized users shall be able to view the details of a case found.
- FR-005.7. If no matches are found, the system shall inform the user that no results were found.
- FR-005.8. The search shall respect the user's access restrictions.

Acceptance Criteria:

1. A search by client name returns matching case files.
2. The results allow users to identify the cases found.
3. A search with no matches displays an informative message.
4. Search results shall not provide access to information for which the user lacks authorization.

3.6. Audit Logging

Identification: FR-006 Name: Modification History Priority: Medium Source: PB-006.

Description:

The system shall record modifications performed by users to maintain a basic traceability history of case files.

Specific Requirements:

- FR-006.1. The system shall identify the user performing a modification.
- FR-006.2. The system shall record the date and time of the modification.
- FR-006.3. Each modification shall generate a new entry in the log.txt file.
- FR-006.4. The system shall append new entries without automatically deleting previous entries.
- FR-006.5. The system shall allow the audit history to be retrieved according to the user's permissions.
- FR-006.6. The system shall record the modification events included in its operations.
- FR-006.7. Each log entry shall identify, at a minimum, the user and the time when the action occurred.

Reference Entry Format:

[2026-10-18 10:30:00] User: jlopez | Action: UPDATE_CASE | Case: CASE-001

The format above is illustrative and may be adjusted during implementation.

Acceptance Criteria:

1. A modification generates an entry in the audit file.
2. The entry includes the user, date, time, and action performed.
3. Previous entries are preserved when new events are recorded.
4. Authorized users can retrieve the history.

3.7. CSV Data Import

Identification: FR-007 Name: Case Import Priority: Medium Source: PB-008.

Description:

The system shall allow users to import structured information from a CSV file and create case objects in memory.

Specific Requirements:

- FR-007.1. The system shall allow users to select or specify a CSV file.
- FR-007.2. The system shall read the file contents.
- FR-007.3. The system shall interpret comma-separated values according to the defined input structure.
- FR-007.4. The system shall identify the fields corresponding to case file data.
- FR-007.5. The system shall create case objects from valid records.

- FR-007.6. The system shall validate mandatory fields in imported records.
- FR-007.7. The system shall report when a file cannot be read or its structure is incorrect.
- FR-007.8. The system shall report the import results.
- FR-007.9. Imported records shall be available in memory for the corresponding operations.

Acceptance Criteria:

1. A valid CSV file produces case objects.
2. Fields are assigned to the corresponding attributes.
3. Incomplete records are not accepted without the required validation.
4. An unreadable file or a file with an incorrect structure produces an informative message.

Limitation: PB-008 specifies the creation of objects in memory. Permanent storage of all imported data shall be verified separately because it is not explicitly established as an acceptance criterion for this requirement.

3.8. Automatic Document Generation

Identification: FR-008 Name: Legal Document Generation Priority: Low Source: PB-009.

Description:

The system shall generate text documents using information previously registered in a case file and a predefined template.

Specific Requirements:

- FR-008.1. The system shall allow users to request document generation for a case.
- FR-008.2. The system shall retrieve the available data from the case object.
- FR-008.3. The system shall use data such as the client's name and the defendant's name when available.
- FR-008.4. The system shall insert the data into a predefined text template.
- FR-008.5. The generated document shall be saved in TXT format.
- FR-008.6. The system shall inform the user whether generation was completed successfully.
- FR-008.7. If the data required to complete the template is missing, the system shall inform the user and avoid generating a document containing incorrect information.

Acceptance Criteria:

1. The system generates a TXT file using the defined template.
2. Client and defendant data are incorporated correctly when available.
3. The document can be located in the configured output directory.

4. If essential data is missing, an informative message is displayed.

3.9. Evidence Registration and Retrieval

Identification: FR-009 Name: Evidence Management Priority: High Source: Use Case Diagram and Class Diagram.

Description:

The system shall support the registration of evidence associated with a case file, including descriptive information and the corresponding file path.

Specific Requirements:

- FR-009.1. The system shall allow users to register evidence.
- FR-009.2. Each piece of evidence shall have an identifier.
- FR-009.3. The record shall include a description, type, file path, and date of incorporation.
- FR-009.4. The evidence shall be associated with a case file.
- FR-009.5. The system shall allow users to retrieve the evidence data and file path.
- FR-009.6. Registration and retrieval shall respect the user's permissions.

Acceptance Criteria:

1. Evidence can be registered with its required information.
2. The evidence is associated with the correct case file.
3. The system allows users to retrieve its details.
4. The file path is stored as a reference.

3.10. User and Permission Administration

Identification: FR-010 Name: Administrative Management of Accounts and Permissions Priority: High Source: PB-001 and Use Case Diagram.

Description:

The system shall allow authorized users to manage accounts, assign roles, and restrict functions according to each profile's responsibilities.

Specific Requirements:

- FR-010.1. The system shall allow user accounts to be managed.
- FR-010.2. The system shall allow roles to be assigned.
- FR-010.3. The system shall allow functions to be restricted to specific profiles.
- FR-010.4. The system shall verify permissions before executing a protected operation.

- FR-010.5. The system shall allow partners to exercise the administrative control defined for their role.

This requirement complements FR-001 and organizes the administrative operations represented separately in the use case diagram.

4. NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements establish the quality conditions LegalSync shall satisfy. The original documents do not define quantitative metrics for all of them; therefore, the following conditions are proposed to complete the specification and must be validated by the team.

4.1. Security

Identification: NFR-001.

- NFR-001.1. The system shall require authentication before allowing access to protected operations.
- NFR-001.2. The system shall implement role-based access control.
- NFR-001.3. The system shall prevent users from performing operations for which they are not authorized.
- NFR-001.4. Passwords shall be hidden during entry.
- NFR-001.5. The production version shall protect credentials using appropriate cryptographic mechanisms.
- NFR-001.6. The product vision establishes the need to protect confidential legal information through encryption.
- NFR-001.7. Access to the audit history shall be restricted to authorized users.

4.2. Information Integrity

Identification: NFR-002.

- NFR-002.1. The system shall validate mandatory fields before registering or modifying case files.
- NFR-002.2. Each case file shall have an identifier that distinguishes it from other records.
- NFR-002.3. Documents, evidence, and deadlines shall maintain their association with the corresponding case file.
- NFR-002.4. Data import shall verify that records comply with the expected structure.
- NFR-002.5. Modification operations shall generate the corresponding audit entries.

4.3. Usability

Identification: NFR-003.

- NFR-003.1. The system shall present understandable options for its main operations.
- NFR-003.2. Error messages shall clearly explain the problem.
- NFR-003.3. Search results shall be presented in a way that facilitates case identification.
- NFR-003.4. Legal deadline alerts shall be visually distinguishable from ordinary messages.
- NFR-003.5. Mandatory fields shall be identified or explained to the user.

4.4. Performance

Identification: NFR-004.

- NFR-004.1. The system shall perform registration, update, and retrieval operations without unnecessary blocking.
- NFR-004.2. The search function shall examine stored data and return matching records.
- NFR-004.3. Deadline calculations shall be performed when the corresponding dates are evaluated or retrieved.
- NFR-004.4. The system shall process CSV files according to the memory capacity available in the execution environment.

The provided documents do not establish a maximum record volume or response-time limits. These values shall be defined through testing and stakeholder validation.

4.5. Maintainability

Identification: NFR-005.

- NFR-005.1. The code shall be organized into classes and responsibilities consistent with object-oriented design.
- NFR-005.2. Authentication, case management, auditing, searching, and deadline operations shall be maintained as distinct responsibilities.
- NFR-005.3. Requirement identifiers shall allow functionalities to be linked to their tests.
- NFR-005.4. Document templates and validation rules shall be modifiable without unnecessarily affecting unrelated modules.

4.6. Portability

Identification: NFR-006.

- NFR-006.1. The overall product vision considers access from computers, tablets, and mobile phones.

- NFR-006.2. The academic implementation shall operate in the execution environment defined by the team.
- NFR-006.3. File paths shall be handled according to the operating system being used.
- NFR-006.4. Full multiplatform compatibility shall be considered an evolution objective unless it is explicitly implemented and tested in this version.

4.7. Traceability and Auditing

Identification: NFR-007.

- NFR-007.1. The system shall allow the user responsible for a recorded modification to be identified.
- NFR-007.2. Audit entries shall include the date and time.
- NFR-007.3. The log shall preserve previous entries when new modifications are added.
- NFR-007.4. When applicable, records shall allow an action to be associated with the corresponding case file.

4.8. Availability and Recovery

Identification: NFR-008.

- NFR-008.1. The system shall report when a required file cannot be accessed.
- NFR-008.2. CSV reading errors shall not result in the silent acceptance of invalid records.
- NFR-008.3. The implementation shall include procedures for preserving data files and audit history.
- NFR-008.4. Backup and recovery strategies shall be defined before the system is used with real legal case files.

The source documents do not establish availability percentages, maximum recovery times, or a specific backup policy. These parameters remain to be defined.

5. EXTERNAL INTERFACE REQUIREMENTS

5.1. User Interface

Identification: INT-001.

The system shall provide an interface that allows users to interact with the available functionalities according to their roles.

The interface shall include:

- Login screen or login option.

- Navigation options based on the user's permissions.
- Forms for registering and updating case files.
- Options for associating documents and evidence.
- Deadline and alert displays.
- Case search functionality.
- Confirmation and error messages.
- Administrative functions for managing users and permissions.
- Options for importing CSV files and generating TXT documents.

For the academic version, the Product Backlog allows a console interface or a basic user interface. A final graphical design has not been specified.

5.2. Local File Interface

Identification: INT-002.

LegalSync shall interact with local files to perform the following operations:

File or Format	Intended Use
Credentials file	User and password validation in the academic implementation
log.txt	Record of modifications performed by users.
CSV	Importing case information.
TXT	Generating legal documents using templates
PDF and Word	References to documents associated with case files.

The system shall inform the user when a required file cannot be read or written.

5.3. Interface with External Devices

Identification: INT-003.

The overall product vision considers access from computers, tablets, and mobile phones, as well as notifications sent to lawyers' devices.

However, the academic Product Backlog does not define an API, notification service, mobile application, or specific communication protocol. Therefore, these integrations shall be technically specified before being incorporated into the implementation.

5.4. Communication Interfaces

Identification: INT-004.

The source documents do not establish communication protocols, web services, external APIs, or integrations with judicial platforms.

The academic version may operate using the local mechanisms defined for its functionalities. If remote access, synchronization, or push notifications are subsequently required, the corresponding protocols, security controls, and connectivity conditions shall be specified.

6. DATA MODEL AND MAIN ENTITIES

This section is based on the classes identified in the UML diagram classesLegalSync (1).vpp. Attributes and operations are described according to the available model, without assuming that every relationship or database constraint has already been implemented.

6.1. User Class

Represents a system user.

Identified Attributes:

- userId: User identifier.
- name: Name.
- username: Username.
- password: Login credential.
- active: Activity indicator.

Identified Operations:

- login(): Log in.
- logout(): Log out.
- changePassword(): Change password.
- hasPermission(): Check permissions.

6.2. Lawyer Class

Represents a lawyer.

Identified Attributes:

- position: Position or job title.
- licenseNumber: Professional license number.

Identified Operations:

- isPartner(): Determine whether the lawyer is a partner.
- restricUserAccess(): Restrict user access.
- getDetails(): Retrieve details.

The operation restricUserAccess() retains the name shown in the original model; it may be normalized during implementation.

6.3. LegalAssistant Class

Represents a legal assistant.

Identified Attribute:

- area: Work area.

Identified Operation:

- updateCaseData(): Update case information.

6.4. CaseFile Class

Represents a legal case file.

Identified Attributes:

- caseId: Case identifier.
- docketNumber: Court case number.
- title: Case title.
- status: Case status.
- openingDate: Opening date.
- client: Associated client.
- defendant: Associated defendant.
- assignedLawyer: Assigned lawyer.
- documents: Related documents.
- evidences: Related evidence.
- deadlines: Associated deadlines.

Identified Operations:

- register(): Register a case file.
- update(): Update a case file.
- validateMandatoryFields(): Validate mandatory fields.
- addDocument(): Associate a document.
- addEvidence(): Associate evidence.
- addDeadline(): Associate a deadline.
- generateDocument(): Generate a document.

- importFromCsv(): Import data from CSV.

6.5. Client Class

Represents a law firm's client.

Identified Attributes:

- clientId: Client identifier.
- name: Name.
- idNumber: Identification number.
- phone: Phone number.
- email: Email address.
- address: Address.

Identified Operations:

- register(): Register a client.
- update(): Update client information.
- getDetails(): Retrieve details.

6.6. Defedant Class

The original model uses the name Defedant; the conventional English spelling is Defendant. To maintain correspondence with the original diagram, the original class name is retained here, and correcting it in the class model is recommended.

Identified Attributes:

- defedantId: Defendant identifier.
- name: Name.
- idNumber: Identification number.
- contactInfo: Contact information.
- address: Address.

Identified Operations:

- register(): Register a defendant.
- update(): Update information.
- getDetails(): Retrieve details.

6.7. Document Class

Represents a document associated with a case file.

Identified Attributes:

- documentId: Document identifier.
- name: Name.
- type: File type.
- filePath: File path.
- uploadDate: Date of incorporation.

Identified Operations:

- register(): Register a document reference.
- getFilePath(): Retrieve the file path.
- saveToFile(): Save information to a file.

6.8. Evidence Class

Represents evidence associated with a case.

Identified Attributes:

- evidenceId: Evidence identifier.
- description: Description.
- type: Evidence type.
- filePath: File path.
- collectionDate: Date of incorporation or collection.

Identified Operations:

- register(): Register evidence.
- getFilePath(): Retrieve the file path.
- getDetails(): Retrieve details.

6.9. Deadline Class

Represents a deadline or legal date that must be monitored.

Identified Attributes:

- deadline: Deadline identifier.
- caseId: Associated case identifier.
- description: Deadline description.
- type: Deadline type.
- targetDate: Target date.
- alertDaysBefore: Number of days before the deadline to trigger an alert.
- completed: Completion indicator.

Identified Operations:

- calculateRemainingDays(): Calculate the remaining days.
- isApproaching(): Determine whether the deadline is approaching.
- generateNotification(): Generate a notification.
- markCompleted(): Mark the deadline as completed.

6.10. Notification Class

Represents a notification addressed to a user.

Identified Attributes:

- notificationId: Notification identifier.
- message: Message.
- createDate: Creation date.
- recipient: Recipient.
- read: Read indicator.

Identified Operations:

- display(): Display the notification.
- markAsRead(): Mark as read.
- getMessage(): Retrieve the message.

6.11. AuditLog Class

Represents the audit log.

Identified Attributes:

- logFile: Log file.
- entries: Audit entries.

Identified Operations:

- registerEntry(): Register an entry.
- writeToFile(): Write to a file.
- getHistory(): Retrieve history.

6.12. SearchEngine Class

Represents the case search component.

Identified Attributes:

- caseFiles: Collection of case files.

- lastQuery: Last search query.

Identified Operations:

- loadData(): Load data.
- searchByClientName(): Search by client name.
- getResults(): Retrieve results.

6.13. General Relationships Between Entities

According to the system's purpose and the identified classes:

- A client may be associated with one or more case files.
- A case file is associated with a client, a defendant, and an assigned lawyer.
- A case file may contain references to multiple documents.
- A case file may include multiple pieces of evidence.
- A case file may have multiple legal deadlines.
- Deadlines may generate notifications addressed to the corresponding users.
- Users who perform modifications generate entries in the audit log.
- The search engine queries the case file collection to retrieve matching records.

These relationships express the expected conceptual model. Exact cardinalities, deletion rules, and integrity constraints shall be confirmed in the final UML diagram and implementation.

7. SYSTEM USE CASES

This section organizes the interactions identified in the usecasediagram.vpp use case diagram.

7.1. UC-001: Log In

Primary Actor: Partner, Lawyer, Junior Lawyer, or Legal Assistant.

Objective: Allow an authorized user to access the system.

Preconditions:

- The system is available.
- The user has a registered account.

Main Flow:

1. The user requests to log in.
2. The system requests the username and password.
3. The user enters the credentials.
4. The system validates the information.

5. The system identifies the user's role and permissions.
6. The system grants access to the authorized functions.

Alternative Flows:

- If the credentials are incorrect, the system reports the error and denies access.
- If the account is inactive, the system shall deny login.

Postcondition: The user can access the functions corresponding to their role.

7.2. UC-002: Register a Case File

Primary Actor: Authorized User.

Objective: Create a legal case file.

Preconditions:

- The user has logged in.
- The user has permission to register cases.

Main Flow:

1. The user selects the registration option.
2. The system requests the case file information.
3. The user enters the required fields.
4. The system validates the mandatory fields.
5. The system registers the case file.
6. The system confirms the result.

Alternative Flow: If mandatory information is missing, the system identifies the incomplete fields and requests correction.

Postcondition: The case file becomes available for retrieval.

7.3. UC-003: Update a Case File

Primary Actor: Authorized User.

Objective: Modify information in an existing case.

Main Flow:

1. The user searches for the case file.
2. The system displays the results.
3. The user selects the case.
4. The system displays its information.

5. The user modifies the permitted fields.
6. The system validates the changes.
7. The system updates the case file.
8. The system records the modification in the audit log.

Postcondition: The changes are saved, and the corresponding audit entry is generated.

7.4. UC-004: Search by Client Name

Primary Actor: Authorized User.

Objective: Locate case files associated with a client.

Main Flow:

1. The user enters the client's name or search criterion.
2. The system searches the stored case files.
3. The system identifies matching records.
4. The system displays the results.
5. The user selects a case file to view its details.

Alternative Flow: If no matches exist, the system informs the user that no results were found.

Postcondition: The user can view the case files they are authorized to access.

7.5. UC-005: Associate a Document or Evidence

Primary Actor: Authorized User.

Objective: Link a document reference to a case file.

Main Flow:

1. The user selects a case file.
2. The system displays the available document-related options.
3. The user enters the information and file path.
4. The system registers the reference.
5. The system associates the reference with the case file.
6. The system confirms the result.

Postcondition: The reference becomes available within the corresponding case file.

7.6. UC-006: Manage Legal Deadlines

Primary Actor: Lawyer or Authorized User.

Objective: Register and monitor legal dates.

Main Flow:

1. The user selects a case file.
2. The system allows the deadline description and date to be entered.
3. The user specifies the deadline information.
4. The system registers the information.
5. The system calculates the remaining days.
6. The system determines whether an alert should be displayed.
7. The system displays the deadline information.

Alternative Flows:

- If the date is invalid, the system requests correction.
- If the deadline is completed, the system identifies it as completed.
- If the deadline is approaching, the system displays the corresponding alert.

Postcondition: The deadline is registered and available for monitoring.

7.7. UC-007: View Audit History

Primary Actor: Authorized User.

Objective: Retrieve recorded modifications.

Main Flow:

1. The user requests to view the history.
2. The system verifies the user's permissions.
3. The system reads the available log entries.
4. The system displays the history.
5. The user reviews the modification details.

Postcondition: The authorized user can identify the recorded actions.

7.8. UC-008: Import Case Files from CSV

Primary Actor: Authorized User.

Objective: Load structured information from a file.

Main Flow:

1. The user specifies the CSV file.
2. The system verifies that the file can be read.
3. The system interprets the records.
4. The system validates the required fields.

5. The system creates the corresponding case objects.
6. The system reports the result of the operation.

Alternative Flow: If the file has an incorrect structure, the system reports the error.

Postcondition: Valid records become available in memory for the corresponding operations.

7.9. UC-009: Generate a Legal Document

Primary Actor: Authorized User.

Objective: Generate a TXT document from case file data.

Main Flow:

1. The user selects a case file.
2. The system retrieves the required information.
3. The user requests document generation.
4. The system completes the template.
5. The system generates the TXT file.
6. The system informs the user where the document was saved.

Alternative Flow: If essential information is missing, the system reports the problem.

Postcondition: The document is generated when the required data is available.

7.10. UC-010: Manage Accounts and Permissions

Primary Actor: Partner.

Objective: Control users and their authorizations.

Main Flow:

1. The partner accesses the administrative module.
2. The system verifies the partner's permissions.
3. The partner selects an account.
4. The partner creates or modifies the account, assigns a role, or restricts functions.
5. The system validates the operation.
6. The system saves the changes.

Postcondition: The account is configured according to the assigned permissions.

8. BUSINESS RULES

The following rules specify the operating conditions necessary for the system to function.

- BR-001. Access Control: Protected operations may only be performed by authorized users.
- BR-002. Case Registration: A case file cannot be registered unless it satisfies the mandatory-field requirements.
- BR-003. Document Association: Each document or piece of evidence must be associated with a case file.
- BR-004. Deadline Monitoring: Remaining days shall be calculated using the target date and the current date.
- BR-005. Alerts: An alert shall be displayed when a deadline falls within the configured advance-notice period.
- BR-006. Auditing: Recorded modifications shall identify the user and the time of the action.
- BR-007. Search: Global search shall primarily use the client's name.
- BR-008. Import: Imported data shall conform to the defined CSV structure.
- BR-009. Document Generation: Documents shall be created using a predefined template and the available data.
- BR-010. Administrative Permissions: Partners may restrict the functions available to other users.
- BR-011. Traceability: Associations between case files, documents, evidence, and deadlines shall be maintained during retrieval and updates.

Rules concerning judicial business days, duplicate clients, document retention, and case deletion require additional validation because they are not sufficiently defined in the provided materials.

9. GENERAL ACCEPTANCE CRITERIA

The LegalSync system shall be considered functionally acceptable when it satisfies the following conditions:

Code	Acceptance Criterion	Related Requirements
AC-001	Users can log in and log out, and the system validates their credentials.	FR-001
AC-002	Partners can manage accounts and permissions according to their authorizations.	FR-001, FR-010

AC-003	Case files can be registered and updated with mandatory-field validation.	FR-002
AC-004	Document and evidence references remain associated with their case files.	FR-003, FR-009
AC-005	Deadlines are registered, calculated, and displayed through alerts when appropriate.	FR-004
AC-006	The search function retrieves case files using the client's name.	FR-005
AC-007	Modifications generate entries in log.txt containing the user, date, and time.	FR-006
AC-008	A valid CSV file allows case objects to be created in memory.	FR-007
AC-009	The system generates TXT documents using the template and available data.	FR-008
AC-010	Protected operations respect the assigned permissions.	FR-001, FR-010

These criteria shall be verified through functional testing before the implementation is considered complete.

10. REQUIREMENTS TRACEABILITY MATRIX

This matrix connects the original Product Backlog functionalities with the requirements defined in this SRS.

Product Backlog	Original Functionality	SRS Requirement	Priority
PB-001	Authentication and role management	FR-001, FR-010	High
PB-002	Case registration and updating	FR-002	High
PB-003	Document reference registration	FR-003	High
PB-004	Local legal deadline alerts	FR-004	High
PB-005	Global search by client	FR-005	High
PB-006	Audit logging	FR-006	Medium
PB-007	Basic login	FR-001	Medium
PB-008	CSV data import	FR-007	Medium
PB-009	Automatic document generation	FR-008	Low

Requirements FR-009 and FR-010 extend the organization of the functionalities identified in the UML diagrams and Product Backlog. FR-009 corresponds to evidence management in the class model, while FR-010 separates permission administration from authentication.

11. REQUIREMENTS OUTSIDE THE CONFIRMED SCOPE

To prevent misunderstandings during development, the following functionalities are identified as appearing in the product vision without their complete implementation being guaranteed by the Product Backlog acceptance criteria.

11.1. Push Notifications to Mobile Phones

The product vision proposes sending proactive notifications to lawyers' phones. However, the academic requirement PB-004 establishes local visual alerts when the system starts.

Therefore, push notifications shall be considered an extension requiring further specification.

11.2. File Upload and Preview

The product vision describes uploading and previewing PDF and Word documents. PB-003, on the other hand, specifies storing file paths as text attributes without analyzing the files' contents.

Actual document viewing shall be validated as a separate functionality.

11.3. Multiplatform Remote Access

The product vision proposes access from different devices and locations. Nevertheless, the academic Product Backlog does not establish a deployment architecture, remote-access infrastructure, or communication protocol.

Full compatibility shall be verified before being declared implemented.

11.4. Data Encryption in Production

Protecting confidential information is identified as a product requirement. However, the documents do not specify a particular encryption algorithm, protocol, or procedure.

These details shall be defined before LegalSync is used with real legal information.

11.5. Bulk Migration of Historical Data and Physical Files

The project definition mentions migrating historical Excel data and digitizing physical files. PB-008 specifies CSV import, but it does not detail a complete migration of all historical records or a document-digitization process.

These activities require additional requirements to establish formats, validation, data retention, and control procedures.

12. IDENTIFIED RISKS

Code	Risk	Possible Consequence	Proposed Mitigation

R-001	Incomplete or incorrect data	Inconsistent case files.	Validate fields and review records.
R-002	Loss of local files	Inability to retrieve information.	Define backup and recovery procedures.
R-003	Unauthorized access	Exposure of confidential information.	Implement authentication and permission controls.
R-004	Errors in legal dates	Incorrect alerts or missed deadlines.	Validate dates and test remaining-days calculations.
R-005	References to nonexistent paths	Difficulty accessing documents.	Verify paths when file access is required.
R-006	Incorrect CSV import	Incorrectly assigned or incomplete records.	Validate file structure, fields, and contents.
R-007	Modification or loss of the audit log	Reduced traceability.	Restrict access and define backup mechanisms.
R-008	Confusion between proposed and implemented functions	Unmet expectations.	Keep the traceability matrix updated.

13. VERIFICATION AND VALIDATION

Verification shall determine whether the system satisfies the established requirements. Validation shall determine whether the functionalities address the legal case management needs identified in HW04.

13.1. Functional Testing

The following tests shall be performed:

1. Log in with valid and invalid credentials.
2. Verify access restrictions for different user roles.
3. Register a complete case file and another with missing mandatory fields.
4. Update a case file and verify the corresponding audit entry.
5. Associate a document reference and verify that it is linked to the case.
6. Register a deadline and verify the remaining-days calculation.
7. Verify that alerts appear for approaching deadlines.
8. Search for case files by client name.
9. Import a valid CSV file and another with an incorrect structure.
10. Generate a TXT document and verify that the data is included.
11. Register evidence and verify its association with the case file.
12. Retrieve the audit history using different user profiles.

13.2. Security Testing

The following conditions shall be verified:

- Incorrect credentials do not grant access.
- Restricted functions cannot be executed by a profile without authorization.
- Passwords are hidden during entry.
- Access to the audit log is controlled.
- Confidential data is protected according to the security level defined for the version.

13.3. Integrity Testing

The following conditions shall be verified:

- Case files satisfy mandatory-field requirements.
- Identifiers distinguish individual records.
- Documents, evidence, and deadlines are associated with the correct case file.
- CSV import does not silently accept invalid records.
- Recorded modifications are consistent with the operations performed.

14. LIMITATIONS AND ITEMS REQUIRING FURTHER DEFINITION

The source documents make it possible to define LegalSync's main functionalities, but they do not contain every parameter necessary for a fully detailed technical specification.

Before implementing or deploying the product, the team shall agree on the following:

1. The programming language and platform version.

2. The exact structure of the credentials and CSV import files.
3. The final mandatory fields for clients, defendants, and case files.
4. The complete permission matrix for the four user roles.
5. The rules for calculating judicial business days and handling overdue dates.
6. The final format of audit entries.
7. The output paths for generated documents.
8. Error handling during file imports.
9. Quantitative performance requirements and expected data volume.
10. Backup, recovery, and information-protection mechanisms.
11. The architecture required for remote access and mobile notifications.
12. Case file retention, deletion, and modification rules.

These aspects shall be documented without altering the requirements that have already been approved.

15. CONCLUSIONS

The LegalSync system proposes a solution to improve the administrative management of legal cases by centralizing case files, clients, defendants, documents, evidence, and procedural deadlines.

This requirements specification establishes the functionalities needed for user authentication, permission management, case registration and updating, document reference association, deadline monitoring, global search, auditing, information import, and document generation.

Defining functional and non-functional requirements facilitates development planning and establishes verifiable criteria for evaluating system behavior. In addition, the traceability matrix connects the functionalities to the original Product Backlog and helps maintain consistency between project needs, UML design, and testing.

Finally, the specification distinguishes the functionalities established for the academic implementation from those proposed as part of the future product vision. This distinction defines the product's scope and prevents features that have not yet been specified from being considered implemented requirements.

LegalSync aims to reduce manual tasks, improve information organization, and provide greater traceability in legal case management while maintaining data integrity, confidentiality, and access control as essential considerations.